

PRD — OnboardAI: AI Employee Pre-boarding & Onboarding Command Center

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Status	Demo / MVP

1. Problem

Onboarding is spread across many tools and people:

- offer and document collection,
- background verification (BGV) and payroll setup,
- policy documents in a shared drive,
- welcome emails and orientation invites sent by hand.

This causes three problems:

- **People Ops time goes into repetitive work.** They answer the same policy questions and chase status by hand.
- **New hires get inconsistent answers.** Generic HR bots answer confidently but not from *Aarohan's* actual policies.
- **Managers can't see what is blocked.** A BGV discrepancy or a missing bank form isn't noticed until payroll is delayed.

2. Objective

Give People Ops and managers one command center that:

1. welcomes each new hire in a personal, consistent way (email + calendar invite for virtual roles),
2. answers policy questions **only** from company policy, citing the source and showing confidence,
3. shows onboarding progress and blockers for each hire, and
4. escalates sensitive topics to a human instead of answering.

Success measures for the demo:

- All 4 MCP tools can be called from the UI.
- 0 secrets in the browser or in the repo.
- Every policy answer shows its source.
- Sensitive questions are always escalated.

3. Users

User	Need
People Ops coordinator (primary)	Welcome hires, track progress, answer policy questions fast
Hiring manager	"Is my new hire ready for Day 1? What's blocking them?"
New hire (secondary, via the assistant)	Quick, trustworthy policy answers

4. User journeys

1. **Welcome a hire.** Dashboard → New Hire Welcome → pick or add a hire → *Generate* (Groq drafts the note) → *Send* (MCP sends a Gmail email and, for virtual roles, a Calendar invite) → the dashboard shows "Sent".
2. **Answer a policy question.** Policy Assistant → type a question or click a chip:
 - *Quick Lookup* returns the exact policy + Drive link.
 - *Policy Intelligence* returns a RAG answer with confidence and source section.
3. **Check progress.** Onboarding Progress → pick a hire → see %, status, pending/completed tasks, blockers and next priorities.
4. **Ask the assistant.** Type "How is EMP-1003 doing?" → Groq calls `get_onboarding_status` → the reply lists blockers → "Tools used" shows the MCP call.
5. **Sensitive question.** "Can I get a salary hike?" → the question is escalated to People Ops and no LLM answer is given.

5. Features & functional requirements

ID	Requirement	Priority
F1	Dashboard cards: new hires, average progress, pending tasks, a policy assistant shortcut, and a new-hire table	Must
F2	Add a new hire (name, email, department, role, joining date, virtual flag). A default task checklist is created.	Must
F3	Generate a personalised welcome note (Groq)	Must
F4	Send the welcome email via MCP. Create a calendar event only if the role is virtual. Record it in Supabase.	Must
F5	Policy Lookup via MCP, with a source link	Must
F6	Policy Intelligence via MCP, with confidence, similarity and human-review flag, plus a Groq plain-English summary	Must
F7	Onboarding progress via MCP, with a Supabase fallback for hires added in the app	Must
F8	Chat assistant using Groq tool-calling over the MCP tools, showing which tools were used	Must
F9	Escalate pay, legal, grievance and termination questions	Must
F10	<code>/api/selftest</code> health check covering all integrations (no side effects)	Should
F11	Proactive day-3 nudge for incomplete tasks	Could (future)

6. Non-functional requirements

- **Security:** keys only in server env vars; no secrets in git; HTML-escaped output; input validation; RLS enabled.
- **Simplicity:** no framework, no build step, zero npm dependencies, one serverless function.
- **Performance:** a single tool call responds in under ~5 seconds; chat in under ~15 seconds.
- **Usability:** plain English, responsive down to phone width, colours taken from the Aarohan logo.
- **Portability:** the n8n workflows are backed up as JSON and a migration path is documented.

7. AI capabilities

Capability	How
Personalised writing	Groq, with a warm-tone system prompt (mirrors the <code>onboardai-warm-tone</code> skill)
Grounded Q&A	Retrieval over Supabase pgvector + deterministic confidence gating. The LLM only rephrases the evidence.

Capability	How
Agentic tool use	A Groq tool-calling loop with 4 tools, up to 4 rounds
Guardrails	A keyword escalation check before the LLM runs, plus system-prompt rules for privacy and "never guess"

8. MCP / tool integrations

MCP server (n8n)	Tool	Side effects
onboardai-policy-lookup	get_policy	none
onboardai-policy-intelligence	answer_policy_question	none
onboardai-onboarding-progress	get_onboarding_status	none
onboardai-new-hire-welcome	onboard_new_hire	sends a Gmail email, creates a Google Calendar event

The Cowork side of the capstone uses the same 4 MCP servers plus Gmail, Google Calendar, Google Drive, Supabase, n8n and Render/Netlify connectors. It also includes the **onboardai-concierge** plugin with 5 skills: policy-answers, warm-tone, escalate-sensitive, progress-summary and employee-data-privacy.

9. Data sources

- **Supabase:** onboarding_employees, onboarding_tasks, onboarding_activity, policy_documents, policy_chunks (38 chunks, 768-dimension vectors).
- **n8n Data Tables:** Employees, Tasks, Documents, Policies (backed up in n8n-backup/data-tables/).
- **Google Drive:** source policy PDFs and markdown, linked in the answers.
- All people and policies are **fictional**.

10. Architecture

Browser (static) → Netlify Function /api/* → Groq · Supabase REST · n8n MCP servers (Streamable HTTP). See the README for the diagram.

11. Security

- GROQ_API_KEY is stored as a Netlify secret env var. The Supabase key is also server-side only.
- The MCP endpoints have no authentication. This is acceptable for fictional data (the assignment allows it). Before any real use, add header auth (MCP_AUTH_HEADER is already supported).
- The RLS demo policies allow anon read and insert. These must be tightened, and Supabase Auth added, before any real data is used.

12. Deployment

Netlify: publish public/, functions in netlify/functions/, no build command. The env vars are listed in the README. After deploying, check /api/selftest.

13. Limitations

- n8n trial expiry (backups are done).
- No authentication or roles.

- Employee data is split between n8n Data Tables and Supabase.
- Gemini is used for embeddings inside n8n (the stored vectors depend on it).
- The welcome email template is fixed in n8n; the Groq-written note is shown in the app.

14. Future roadmap

1. **Week 1:** Move the 4 tools to Supabase Edge Functions and a single data store.
2. **Week 2:** Supabase Auth with roles (new hire sees only their own record).
3. **Week 3:** Day-3 proactive Slack/email nudges; manager weekly digest (progress-summary skill).
4. **Later:** Document upload with verification status; buddy assignment in Slack; analytics on time-to-productive.